

## 1 · Data

**Public dataset**

Nature 2025 · MIT licence

341 formulations

6 monomers on the  
composition simplex

Target: adhesion  
strength (kPa)

## 2 · Training region

**Training set**

n = 180 · low-performance

5-fold grouped CV

5 seeds · 25 models

Ablation-gated  
components

## 3 · SIMPLEX

**SIMPLEX**

dual-modality encoder

Monomers +  
pairwise terms

ResBlock x2 +  
interaction attention

Mixup · SWA ·  
domain constraint

## 4 · Extrapolation

**External cohort**

n = 161 · SMBO-discovered

High-performance  
composition region

Target-value shift  
(mean 47 → 154 kPa)

Evaluated once,  
after freezing

## 5 · Screening

**Ranking + insight**

Spearman  $\rho$  = 0.50  
vs RF 0.21

Top-20 precision 0.25  
vs RF 0.10

Permutation importance  
→ composition synergy

**Accelerated screening**